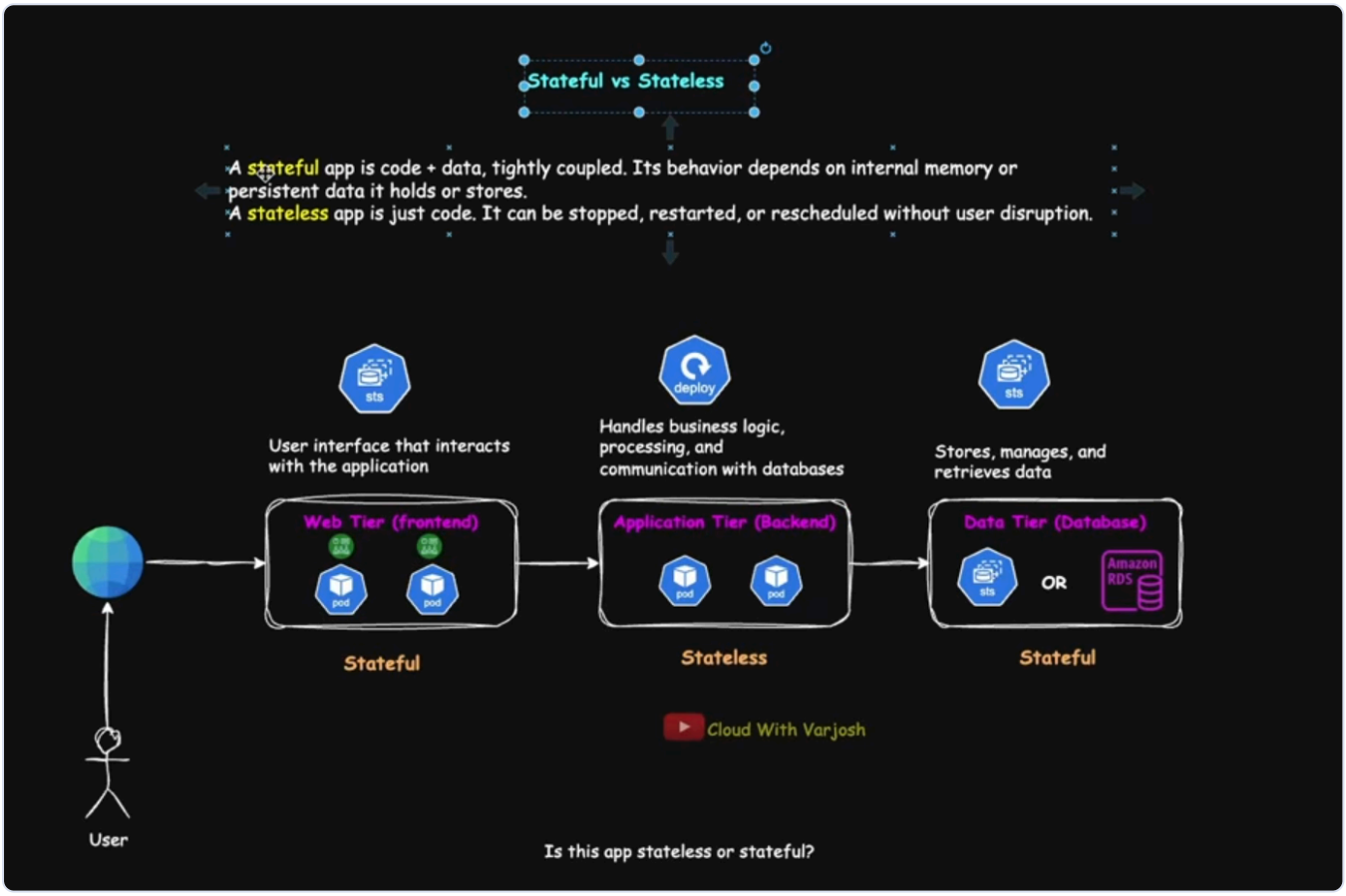
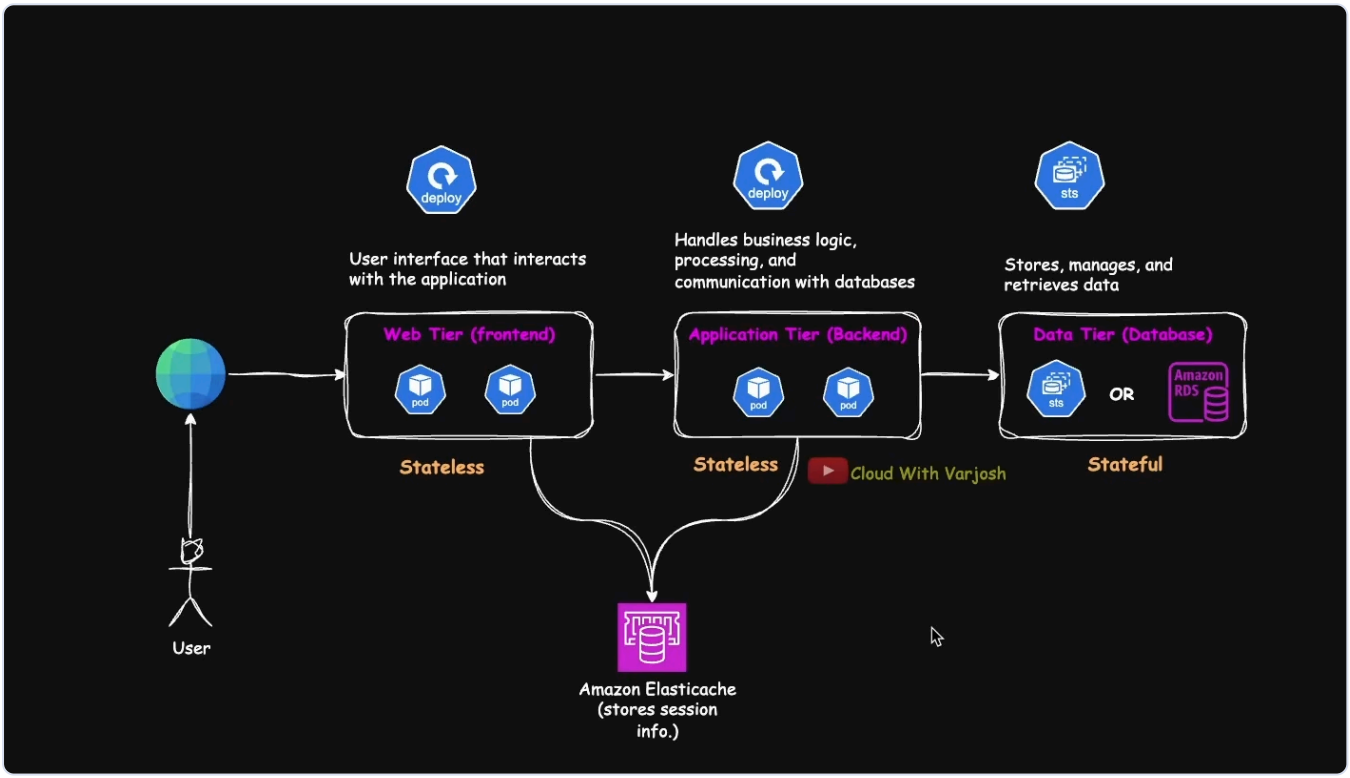
KUBERNETES PREPARATION · MY LEARNING NOTES · SESSION 10

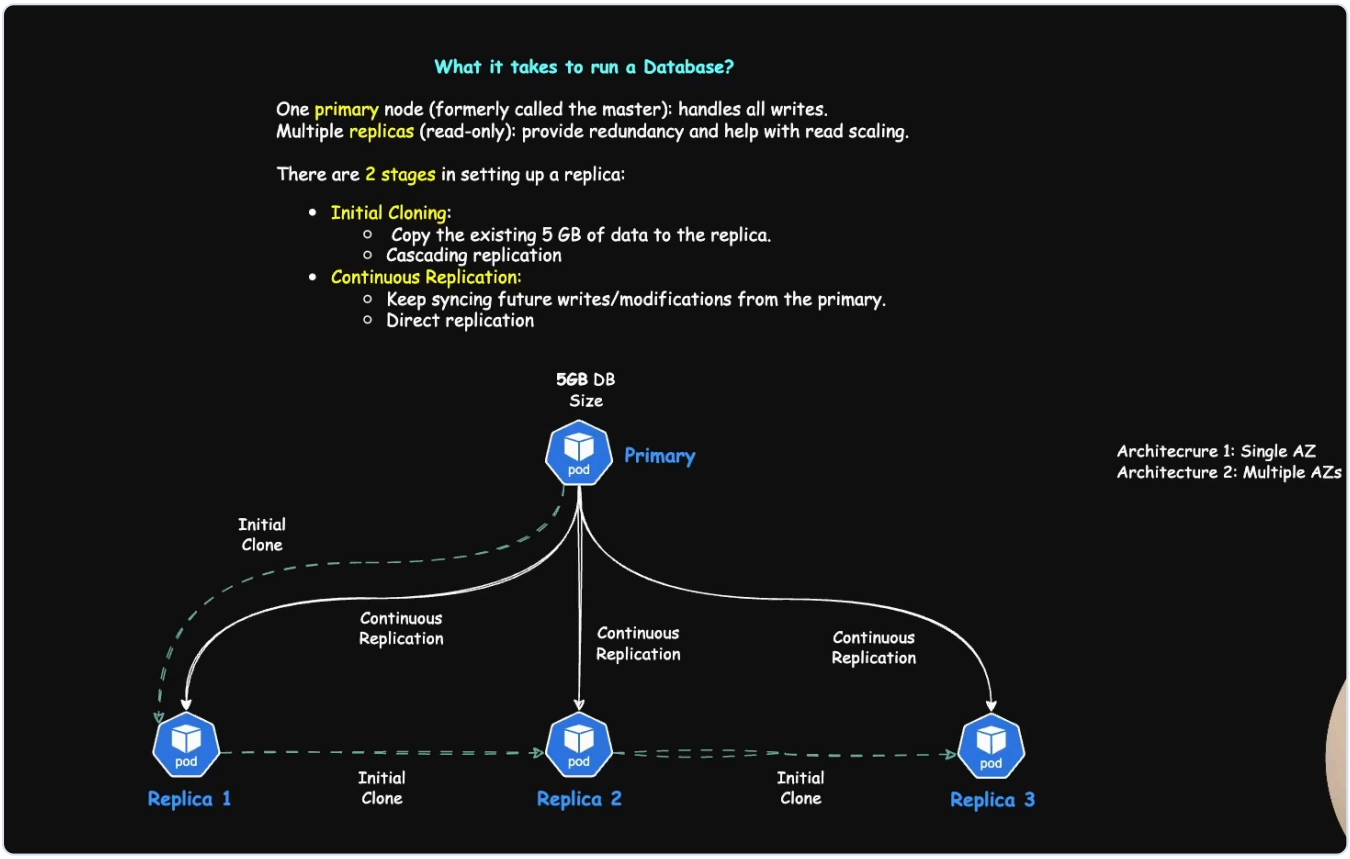
StatefulSets — Concepts

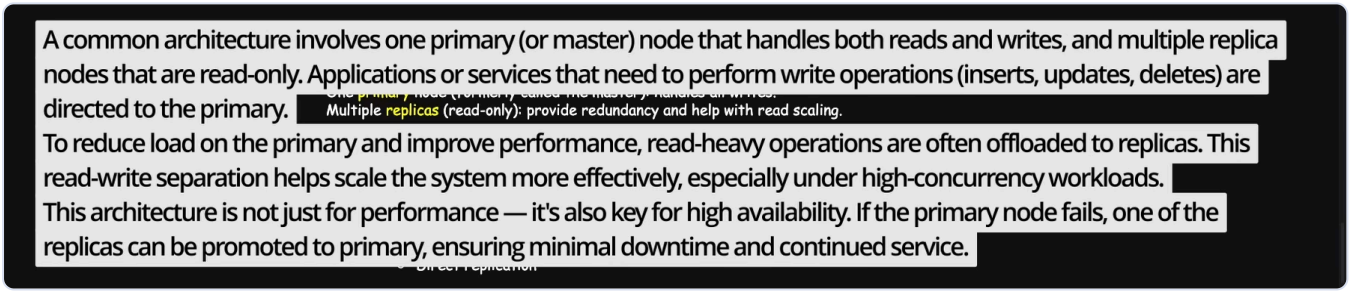
Screenshot gallery — 9 captures, in session order

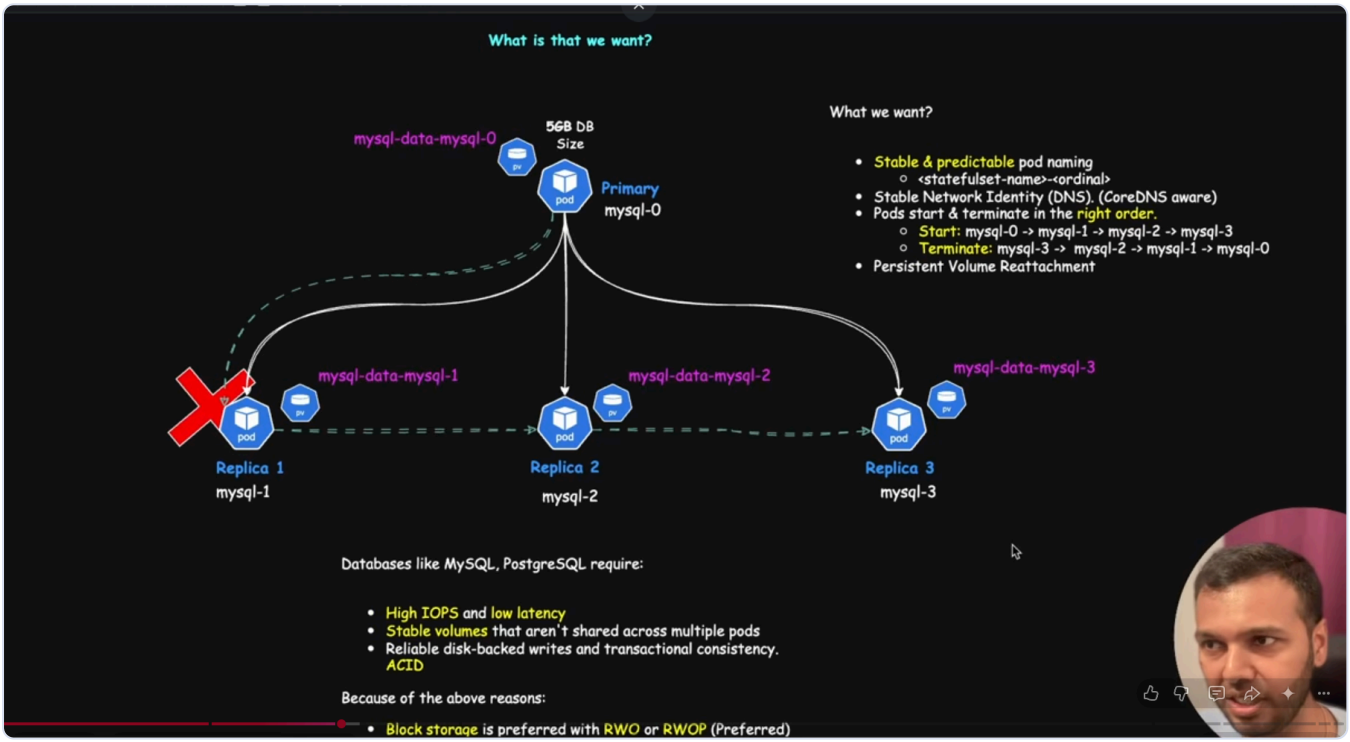

Agenda:
What are Stateful & Stateless apps
Why & What of StatefulSets
Demo 1: Implementing StatefulSets on Amazon EKS
Demo 2: Chaos Engineering: Volume Reattachment

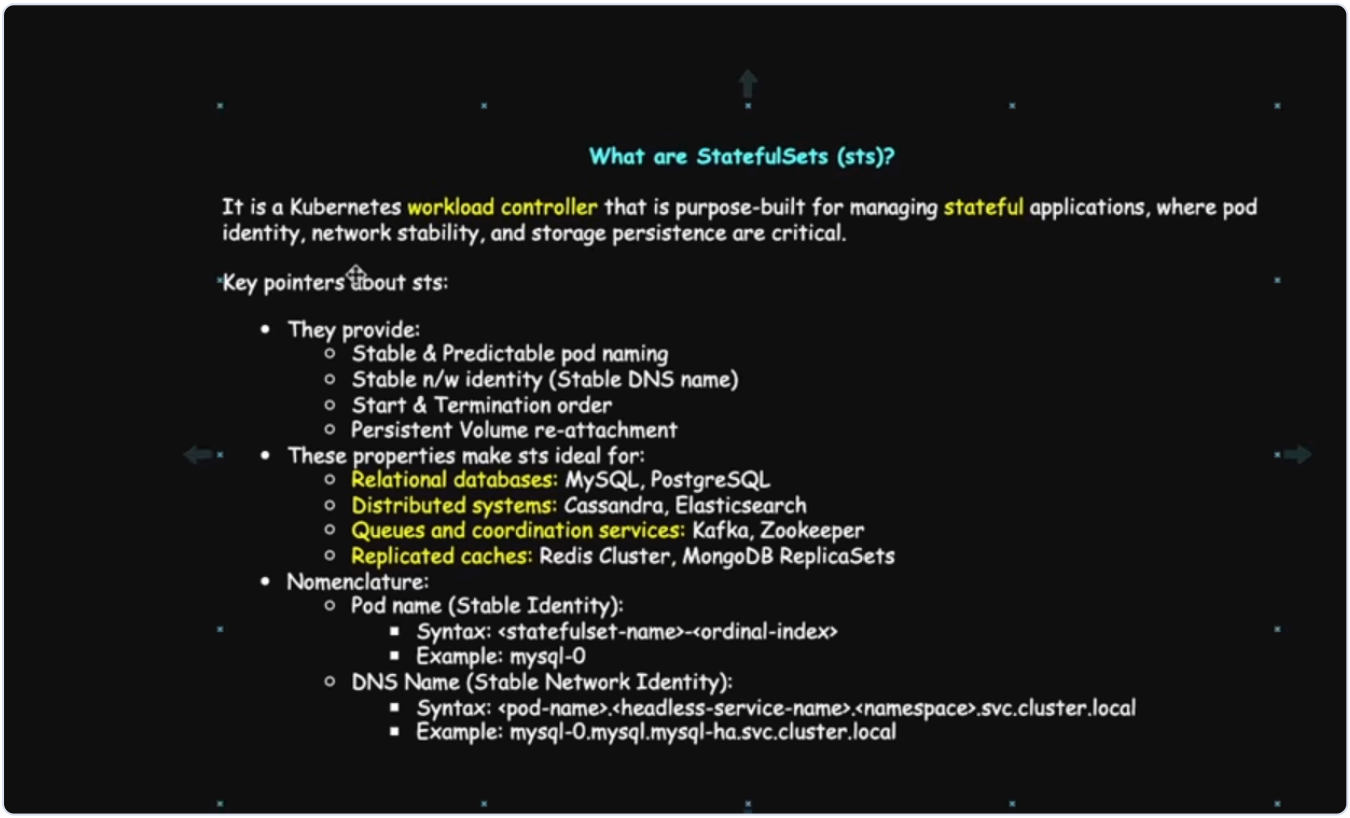

Stateful vs Stateless
A stateful app is code + data, tightly coupled. Its behavior depends on internal memory or persistent data it holds or stores.
A stateless app is just code. It can be stopped, restarted, or rescheduled without user disruption.
sts
User interface that interacts with the application
deploy
Handles business logic, processing, and communication with databases
sts
Stores, manages, and retrieves data
Web Tier (frontend)
pod
pod
Application Tier (Backend)
pod
pod
Data Tier (Database)
sts
OR
Amazon RDS
Stateful
Stateless
Stateful
Cloud With Varjosh
User
Is this app stateless or stateful?

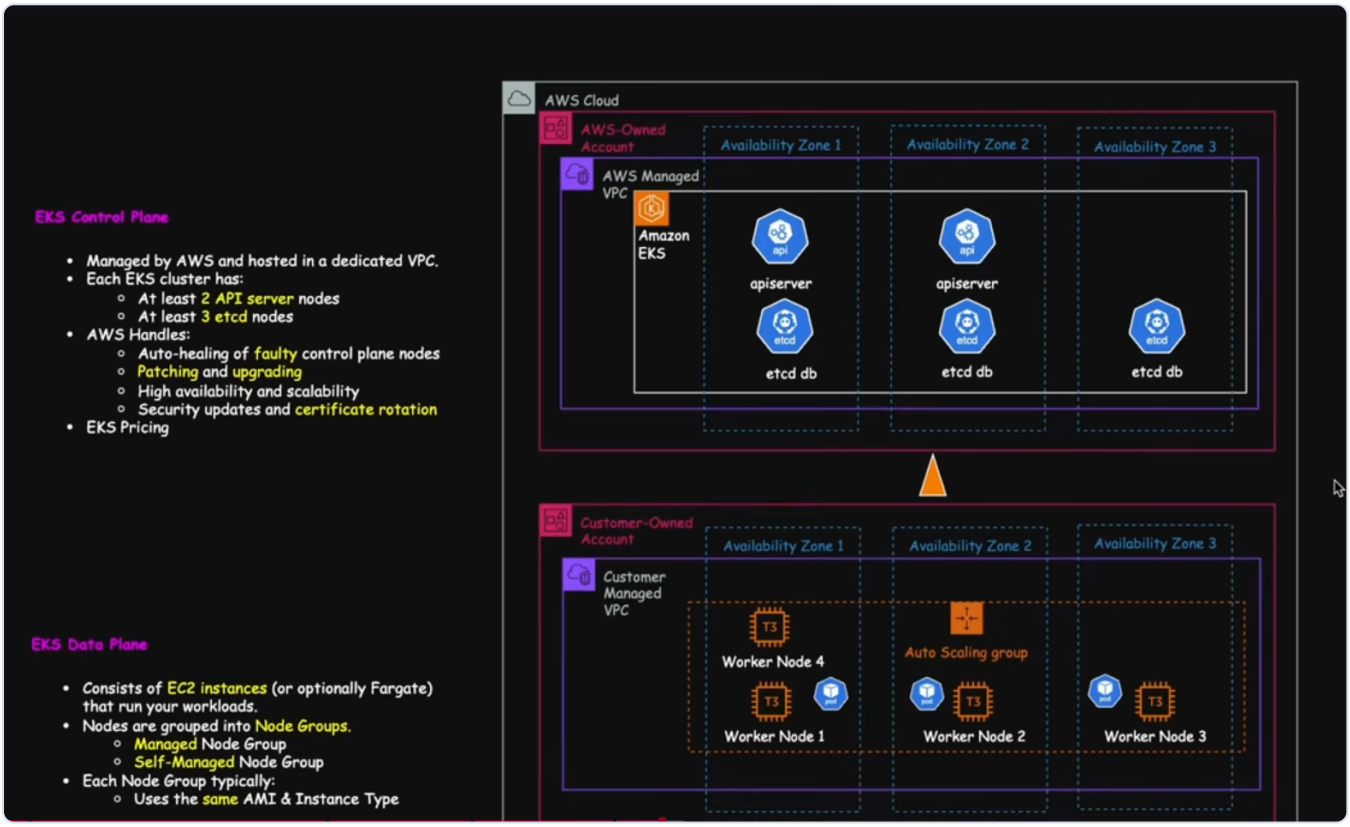

deploy
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sts
User interface that interacts with the application
Handles business logic, processing, and communication with databases
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Web Tier (frontend)
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Data Tier (Database)
pod
pod
pod
pod
sts
OR
Amazon RDS
Stateless
Stateless
Cloud With Varjosh
Stateful
User
Amazon Elasticache (stores session info.)

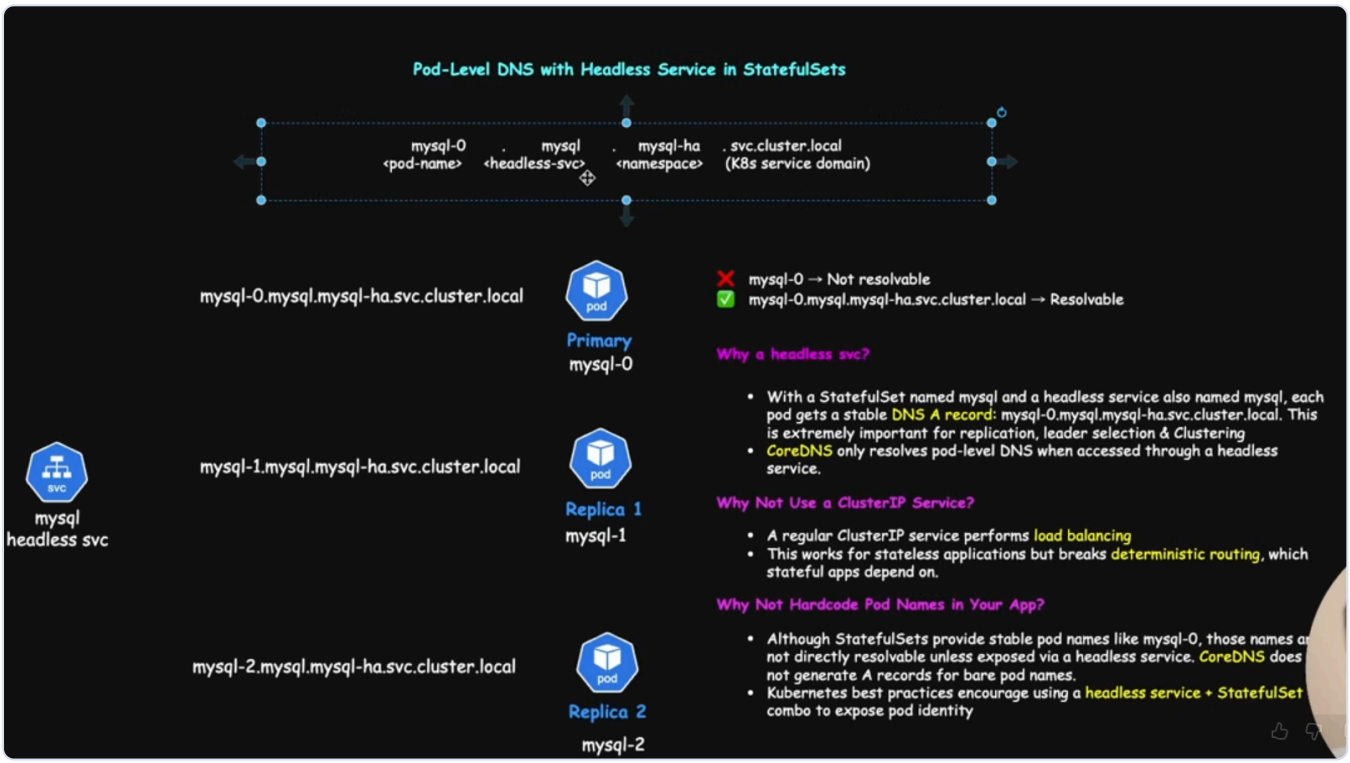

What it takes to run a Database?
One primary node (formerly called the master): handles all writes.
Multiple replicas (read-only): provide redundancy and help with read scaling.
There are 2 stages in setting up a replica:
Initial Cloning:
Copy the existing 5 GB of data to the replica.
Cascading replication
Continuous Replication:
Keep syncing future writes/modifications from the primary.
Direct replication
5GB DB Size
pod
Primary
Initial Clone
Continuous Replication
Continuous Replication
Continuous Replication
pod
Replica 1
Initial Clone
pod
Replica 2
Initial Clone
pod
Replica 3
Architecrure 1: Single AZ
Architecture 2: Multiple AZs


A common architecture involves one primary (or master) node that handles both reads and writes, and multiple replica nodes that are read-only. Applications or services that need to perform write operations (inserts, updates, deletes) are directed to the primary.
Multiple replicas (read-only): provide redundancy and help with read scaling.
To reduce load on the primary and improve performance, read-heavy operations are often offloaded to replicas. This read-write separation helps scale the system more effectively, especially under high-concurrency workloads.
This architecture is not just for performance — it's also key for high availability. If the primary node fails, one of the replicas can be promoted to primary, ensuring minimal downtime and continued service.


What is that we want?
mysql-data-mysql-0
5GB DB Size
Primary
mysql-0
What we want?
Stable & predictable pod naming
<statefulset-name>-<ordinal>
Stable Network Identity (DNS). (CoreDNS aware)
Pods start & terminate in the right order.
Start: mysql-0 -> mysql-1 -> mysql-2 -> mysql-3
Terminate: mysql-3 -> mysql-2 -> mysql-1 -> mysql-0
Persistent Volume Reattachment
mysql-data-mysql-1
mysql-data-mysql-2
mysql-data-mysql-3
Replica 1
mysql-1
Replica 2
mysql-2
Replica 3
mysql-3
Databases like MySQL, PostgreSQL require:
High IOPS and low latency
Stable volumes that aren't shared across multiple pods
Reliable disk-backed writes and transactional consistency.
ACID
Because of the above reasons:
Block storage is preferred with RWO or RWOP (Preferred)


What are StatefulSets (sts)?
It is a Kubernetes workload controller that is purpose-built for managing stateful applications, where pod identity, network stability, and storage persistence are critical.
Key pointers about sts:
They provide:
Stable & Predictable pod naming
Stable n/w identity (Stable DNS name)
Start & Termination order
Persistent Volume re-attachment
These properties make sts ideal for:
Relational databases: MySQL, PostgreSQL
Distributed systems: Cassandra, Elasticsearch
Queues and coordination services: Kafka, Zookeeper
Replicated caches: Redis Cluster, MongoDB ReplicaSets
Nomenclature:
Pod name (Stable Identity):
Syntax: <statefulset-name>-<ordinal-index>
Example: mysql-0
DNS Name (Stable Network Identity):
Syntax: <pod-name>.<headless-service-name>.<namespace>.svc.cluster.local
Example: mysql-0.mysql.mysql-ha.svc.cluster.local


EKS Control Plane
Managed by AWS and hosted in a dedicated VPC.
Each EKS cluster has:
At least 2 API server nodes
At least 3 etcd nodes
AWS Handles:
Auto-healing of faulty control plane nodes
Patching and upgrading
High availability and scalability
Security updates and certificate rotation
EKS Pricing
EKS Data Plane
Consists of EC2 instances (or optionally Fargate) that run your workloads.
Nodes are grouped into Node Groups.
Managed Node Group
Self-Managed Node Group
Each Node Group typically:
Uses the same AMI & Instance Type
AWS Cloud
AWS-Owned Account
AWS Managed VPC
Amazon EKS
Availability Zone 1
Availability Zone 2
Availability Zone 3
apiserver
apiserver
etcd db
etcd db
etcd db
Customer-Owned Account
Customer Managed VPC
Availability Zone 1
Availability Zone 2
Availability Zone 3
Worker Node 4
Auto Scaling group
Worker Node 1
Worker Node 2
Worker Node 3


Pod-Level DNS with Headless Service in StatefulSets
mysql-0 <pod-name> . mysql <headless-svc> . mysql-ha <namespace> . svc.cluster.local (K8s service domain)
mysql-0.mysql.mysql-ha.svc.cluster.local
Primary
mysql-0
mysql-1.mysql.mysql-ha.svc.cluster.local
Replica 1
mysql-1
mysql-2.mysql.mysql-ha.svc.cluster.local
Replica 2
mysql-2
svc
mysql
headless svc
pod
mysql-0 → Not resolvable
mysql-0.mysql.mysql-ha.svc.cluster.local → Resolvable
Why a headless svc?
With a StatefulSet named mysql and a headless service also named mysql, each pod gets a stable DNS A record: mysql-0.mysql.mysql-ha.svc.cluster.local. This is extremely important for replication, leader selection & Clustering
CoreDNS only resolves pod-level DNS when accessed through a headless service.
Why Not Use a ClusterIP Service?
A regular ClusterIP service performs load balancing
This works for stateless applications but breaks deterministic routing, which stateful apps depend on.
Why Not Hardcode Pod Names in Your App?
Although StatefulSets provide stable pod names like mysql-0, those names a
not directly resolvable unless exposed via a headless service. CoreDNS does
not generate A records for bare pod names.
Kubernetes best practices encourage using a headless service + StatefulSet
combo to expose pod identity